

Question:

A silicious rock contains the mineral ZnS. To analyze for Zn, a sample of the rock is pulverized and treated with HCl to dissolve the ZnS (silicious matter is insoluable). Zinc is precipitated from solution by the addition of potassium ferrocyanide K₄ Fe (CN)₆. After filtering, the precipitate is dried and weighed. The reactions which occur are $\text{ZnS} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\text{S}$ $2\text{ZnCl}_2 + \text{K}_4\text{Fe}(\text{CN})_6 \rightarrow \text{Zn}_2\text{Fe}(\text{CN})_6 + 4\text{KCl}$ If a 2 gram sample of rock yields 0.969 gram of Zn₂Fe(CN)₆, what is the percentage of Zn in the sample? Atomic weight Zn = 65.4, molecular weight Zn₂ Fe (CN)₆ = 342.6.

Answer Choices:

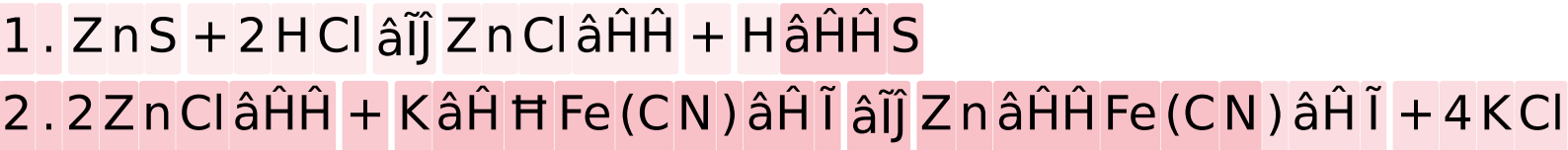
- A. 10 % Zn
- B. 22 % Zn
- C. 20 % Zn
- D. 19 % Zn
- E. 15 % Zn
- F. 18 % Zn
- G. 16 % Zn
- H. 14 % Zn
- I. 12 % Zn
- J. 25 % Zn

Reasoning + Answer:

< reasoning process here >

We are given :

- A 2 gram sample of rock yields 0.969 g of Zn₂Fe(CN)₆.
- We are to find the percentage of Zn in the original rock sample.
- The relevant reactions :



From the stoichiometry :

- 2 moles of ZnCl₂ produce 1 mole of Zn₂Fe(CN)₆.
- Each mole of Zn₂Fe(CN)₆ contains 2 moles of Zn.

We are given :

- Molar mass of Zn₂Fe(CN)₆ = 342.6 g/mol
- Atomic mass of Zn = 65.4 g/mol $\rightarrow$ so 2 moles of Zn = $2 \times 65.4 = 130.8$ g

Step 1 : Find moles of Zn₂Fe(CN)₆ produced
Moles = mass / molar mass = 0.969 g / 342.6 g/mol $\approx$ 0.002829 mol

Step 2 : Each mole of Zn₂Fe(CN)₆ comes from 2 moles of Zn
So moles of Zn = $2 \times 0.002829 = 0.005658$ mol

Step 3 : Mass of Zn = moles $\times$ atomic mass = 0.005658 $\times$ 65.4 $\approx$ 0.370 g

Step 4 : Percentage of Zn in the 2 g sample :
(0.370 g / 2 g) $\times$ 100 = 18.5 % $\approx$ 18 %

So the percentage of Zn is approximately 18 %.

Answer choice : F . 18 % Zn

</reasoning process >
< answer>F</answer>